

Technical write-up — Nova trade-doc prototype

A take-home prototype for GoComet's Nova Full-Stack AI Engineer role. Three agents over LangGraph, OpenAI-compatible LLM transport, single-page HTML UI. **Runs end-to-end on a laptop in under five minutes.** All numbers in this document are measured, not projected, unless explicitly tagged.

Read this together with [PRD.md](#) (product framing) and the ADRs in [docs/adr/](#) (one short rationale per defensible choice).

1 · Problem & scope

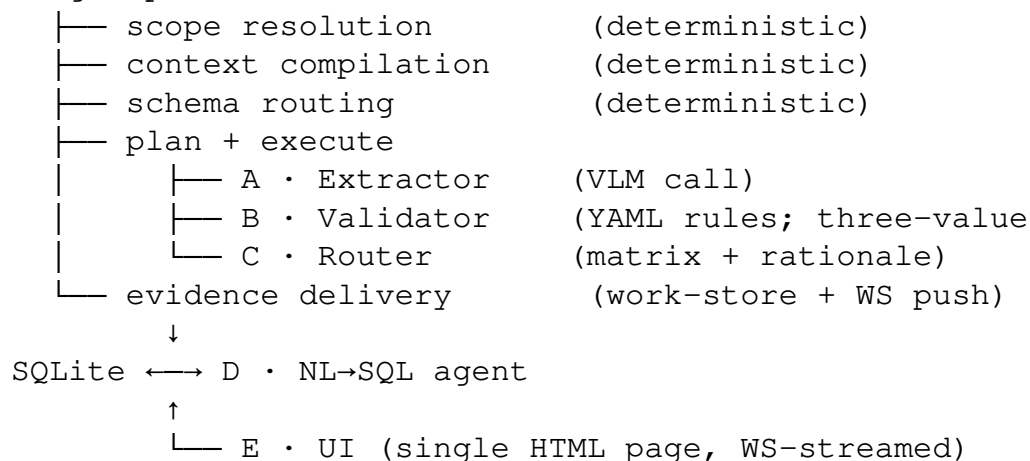
A trade-document operator at ACME Pharma receives Bills of Lading from carriers, must verify eight required fields against their internal rules (approved consignees, HS codes, ports, incoterms, gross-weight format, invoice format), and either auto-approves the declaration, escalates to human review, or drafts an amendment for the carrier.

The prototype demonstrates the five behaviours (A–E) the brief requires on real input, end-to-end, with a single click in the UI per scenario.

2 · Architecture in one paragraph

A FastAPI HTTP/WebSocket surface accepts a PDF and writes it to disk. A LangGraph StateGraph then walks five stages — scope resolution, context compilation, schema routing, plan + execute, evidence delivery — that mirror Nova's runtime. Stage 4 is where the three agents live: an **Extractor** (multimodal VLM call, 8 typed fields with per-field confidence and supporting quote), a **Validator** (deterministic YAML rule engine, three-valued status: match / mismatch / uncertain), and a **Router** (decision matrix that maps the validation table to one of {auto_approve, human_review, draft_amendment} plus an LLM-written rationale). Every artefact persists to a SQLite work-store; an NL→SQL agent answers operator questions over that store. The UI streams every stage event over WebSocket so the operator watches the work happen.

PDF → [FastAPI] → LangGraph



3 · Why the model is swappable

The LLM is reached over the OpenAI-compatible `chat.completions` API with `response_format=json_object` and the `image_url` content channel — both natively supported by vLLM (self-hosted) and OpenAI (hosted). Flipping `LLM_PROVIDER` in `.env` switches the entire pipeline between:

- Qwen/Qwen2.5-VL-7B-Instruct-AWQ (dense, AWQ-INT4, ~5 GB weights)
- cyankiwi/Qwen3.6-35B-A3B-AWQ-4bit (35B-MoE / 3B-active, AWQ-INT4, ~17 GB weights)
- gpt-4o-mini (OpenAI hosted, assessor-friendly)

No code changes. See [docs/MODEL_COMPARISON.md](#) for the side-by-side and the **calibration result** below.

4 · The five behaviours, with file pointers

Behaviour	Implementation
A Multimodal extraction with quotes	<code>src/nova/agents/extractor.py</code>
B Three-valued deterministic validation	<code>src/nova/agents/validator.py</code> + <code>src/nova/schemas/customers/acme/rules.yaml</code>
C Decision routing + rationale	<code>src/nova/agents/router.py</code> + <code>decision.yaml</code>
D NL→SQL over the work-store	<code>src/nova/store/nl_query.py</code>
E Streaming UI (single HTML page)	<code>src/nova/api/static/index.html</code> + <code>src/nova/api/main.py</code>

5 · The "never silently approve" property

The expensive failure in trade documents is a wrong `match` that lets a bad declaration through to customs. Multiple layers exist explicitly to make that failure unreachable:

1. **Closed Pydantic schema.** The model cannot invent new field names — schema validation rejects them.
2. **Quote channel.** Every extracted field must come with a quote. The Extractor verifies each quote against the PDF text layer using exact substring OR a tolerant `difflib.SequenceMatcher` ratio ≥ 0.85 . Missing/weak matches demote confidence to ≤ 0.55 .
3. **System-policy confidence cap on unverifiable extractions.** When there is no text layer to verify against (image-only scan), every field's confidence is capped at `confidence_accept - 0.05 = 0.90`, below the 0.95 accept threshold. Effect: 8/8 rules become uncertain and route to `human_review`.
4. **Three-valued validator.** Any rule whose underlying field is below the 0.95 accept threshold is forced to `uncertain` regardless of whether the value matches the rule.
5. **Router decision matrix.** Any uncertain rule $\rightarrow$ `human_review`. Any high-severity mismatch $\rightarrow$ `human_review`. `auto_approve` requires every rule to come back match.

See [ADR 005](#) for the policy discussion and the model-miscalibration finding that motivated layer (3).

6 · An empirical finding — model calibration changes the picture

The synthetic uncertain BoL is generated with an opaque rectangle stamped over the gross-weight digits, then flattened to image-only PDF so there is no text layer to leak ground truth.

- **Qwen2.5-VL-7B-Instruct-AWQ** returned '1240.5 KG' at **0.97 confidence** — it pattern-matched the field label and surrounding context to invent a plausible number. Even strengthening the system prompt with "do not guess at obscured characters" reduced the reported confidence by only ~ 0.02 . This is a model-level miscalibration, not a prompting problem.
- **Qwen3.6-35B-A3B-AWQ-4bit** returned ' KG' at **0.40 confidence** on the same input — it correctly recognised that the digits were gone.

The system-policy cap (layer 3 above) was added in response to the 2.5-VL-7B result so that the architectural guarantee holds *regardless* of which model is underneath. The 3.6-35B result is an upside — fewer false escalations — not a precondition.

7 · Measured end-to-end numbers

Against the production-target model (cyankiwi/Qwen3.6-35B-A3B-AWQ-4bit) on a single-GPU vLLM endpoint:

Variant	Decision	Cost	Latency	Notes
clean	auto_approve	\$0.0050	5.97 s	8/8 match at 0.98
mismatch	human_review	\$0.0050	4.37 s	HS 8471.30 flagged, high-severity discrepancy
uncertain	human_review	\$0.0051	6.65 s	8/8 uncertain (model conf 0.40 + system cap 0.90)

Cost budget per shipment is \$0.50 $\rightarrow$ we run $\sim 100 \times$ **under budget**. Tests: **14/14 passing** (validator, router, NL $\rightarrow$ SQL safety, synth).

8 · Stack and ADRs

- **Orchestration:** LangGraph StateGraph with AsyncSqliteSaver checkpointer — see [ADR 001](#). 27 checkpoint rows + 102 write rows persisted across the 3-shipment run.
- **3 agents, not 1:** [ADR 002](#).
- **OpenAI-compatible transport:** [ADR 003](#).
- **SQLite now, ClickHouse later:** [ADR 004](#).
- **Confidence + three-valued status:** [ADR 005](#).
- **System-of-Outcomes substrate:** [ADR 006](#).

9 · Operator surface (NL $\rightarrow$ SQL)

Four real example queries with the generated SQL and the rows returned are in

[SAMPLE_QUERIES.md](#). Safety properties: `PRAGMA query_only=1` at the connection level + DDL/DML keyword lint before execution. 3/4 queries returned correct answers in <500 ms; the 4th was a *recall* failure (wrong JSON path) not a *safety* failure, and is fixed by adding one few-shot example.

10 · What this prototype is NOT

- No auth, no multi-tenancy, no real email send.
- Bbox highlighting uses quote-text matching, not pixel-accurate VLM bboxes.
- Eval harness runs on synthesised docs; production path would use a 500-doc human-labelled golden set with isotonic-regression confidence calibration.
- Observability is `structlog` + per-call cost ledger. No LangSmith/Langfuse.

These are deliberate scope cuts. PRD §12 lays out the production path.

11 · Where to look next

- Watch the demo: [DEMO.md](#) — 5-minute walkthrough mapping each click in the UI to the brief's A–E behaviours.
- Read the PRD: [PRD.md](#) — full product framing in Nova/FDE/System-of-Outcomes terms.
- Run it: see `README.md` §"Laptop setup".